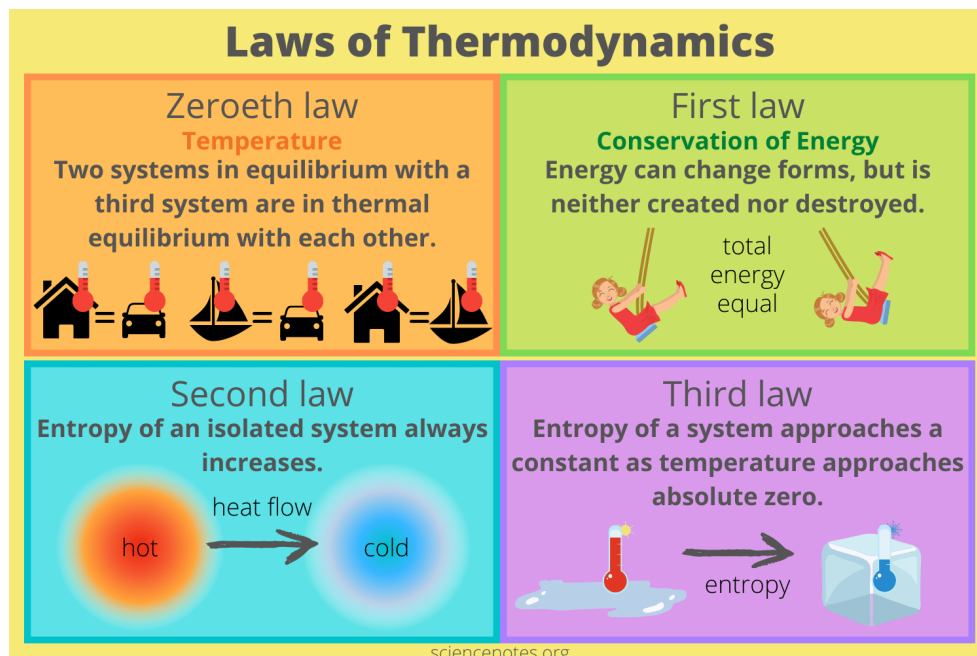


APS110 – Lecture 31

See hand written notes for drawings.

Spontaneous

- In a thermodynamic sense, this has to do with rxns that will proceed on their own
- In thermodynamics, 'spontaneous' does not dictate speed of rxn in any way (recall diamond decomposing into graphite – takes a long time)
- $\text{Na}_{(s)} + \frac{1}{2} \text{Cl}_{2(g)} \rightarrow \text{NaCl}_{(s)}$
 - Spontaneous rxn
 - Exothermic rxn
- Heat engine
 - Every heat engine has a hot source and a heat sink
 - The engine will operate as long as there is a temperature difference (ΔT)
- Thermodynamics, in a way, tells us about the direction of time
 - Hot coffee when left in a room will transfer its heat to the room until the coffee and the room both reach equilibrium
 - We wouldn't expect a cold cup of coffee to spontaneously become hot when left in a room
- Energy cannot be created or destroyed, so you can always apply an energy balance (equilibrium)
 - This is the **first law of thermodynamics**
- There is a directionality to the flow of energy, always flows from high temp to low temp
 - Heat from the room is not going to flow into the cold coffee spontaneously
 - **Second law of thermodynamics** → $\Delta S > 0$
- ***When doing anything with thermodynamics, **temperature is ALWAYS IN KELVIN (K)!!**



- Proteins only form in our body is we eat

- Humans are ordered beings, we are fighting entropy to exist in such an ordered way
- The only way we can do this is to continue eating
- Gibbs energy is a quantity used a lot in solid state chemistry
 - We can say that the Gibbs energy for a given reaction *decreases* meaning the reaction progresses spontaneously
 - But, this will always be related to the total change in entropy of the universe, which **ALWAYS INCREASES**
- Sometimes entropy is described as 'disorder'
 - This is not a very refined description
- A better description is: **entropy is a measure of the uncertainty in knowing the energy state of a randomly selected particle**
 - Recall Boltzmann energy distribution:
 - At 0K the only energy state that is occupied is the lowest energy state
 - This is the only possible way that we can have 'perfect order' (0 entropy)
→ you know exactly where every atom is located, since they are all in the same energy state
 - At infinitely high temperature, all energy states are equally occupied
 - When you randomly select an atom, it is impossible to know where it would be located, since there is an equal chance it came from every energy state
 - Highest entropy
-
-
-
- k